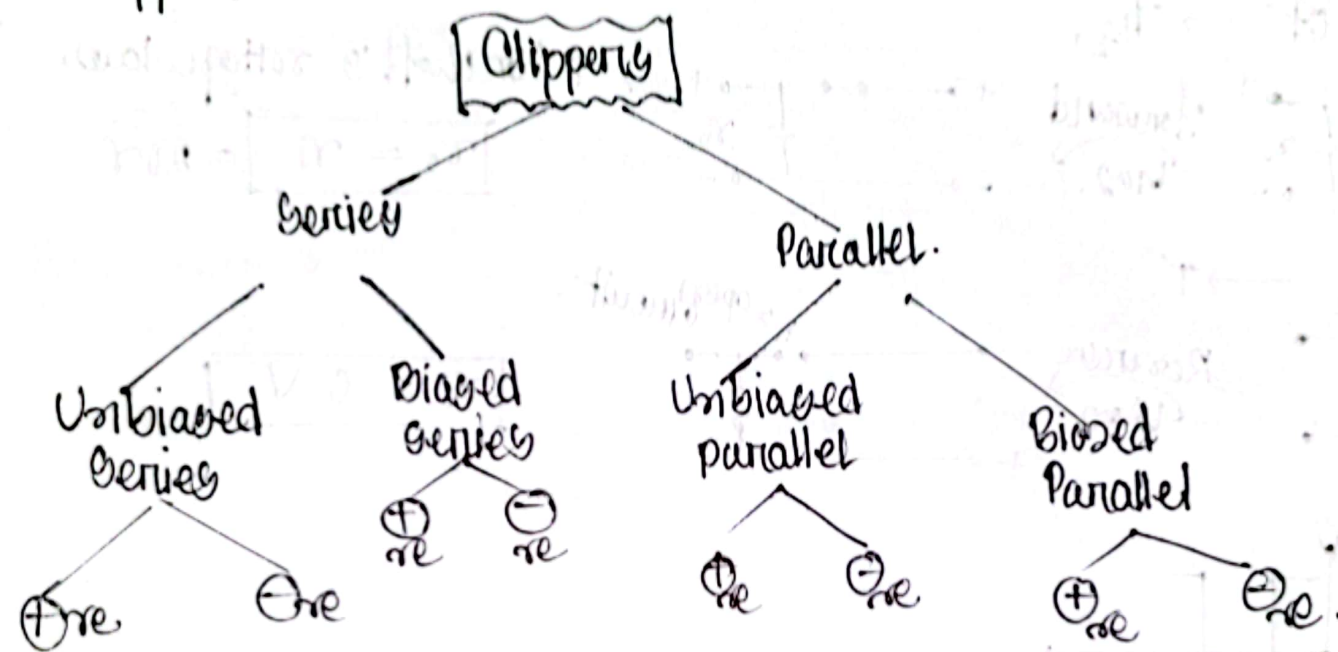


CLIPPERS :-

* Clippers are networks that use diodes to clip a portion of input signal without distorting the remaining part of the waveform.

* If we want to clip portion of the waveform we use clippers.

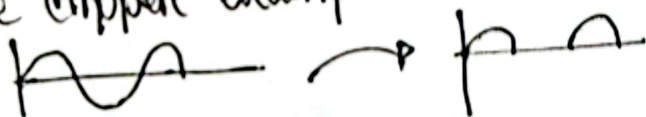


* Positive clipper $\rightarrow$ positive portion of the waveform is clipped.

* Negative clipper $\rightarrow$ negative portion of the waveform will be clipped.

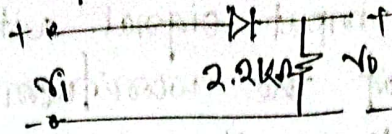
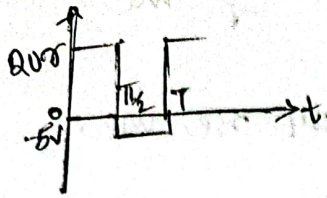
* Biased $\rightarrow$ Basic, Unbiased $\rightarrow$ There is an additional dc source.
There is no additional dc source.

Negative clipper example:

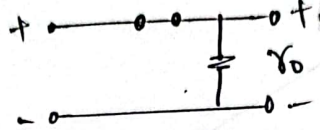
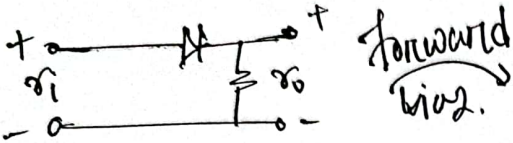


Unbiased Series Clippers:

1. Determine r_o for the input shown.



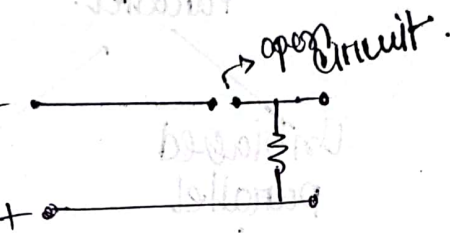
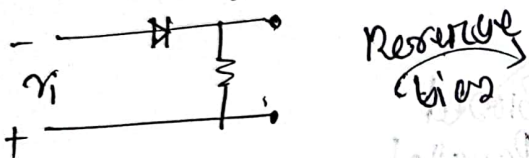
Ans: For $0 < t < T/2$,



Kirchoff's voltage law,

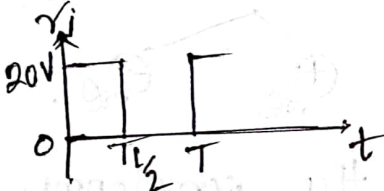
$$v_o = v_i = 20V$$

For $T/2 < t < T$,



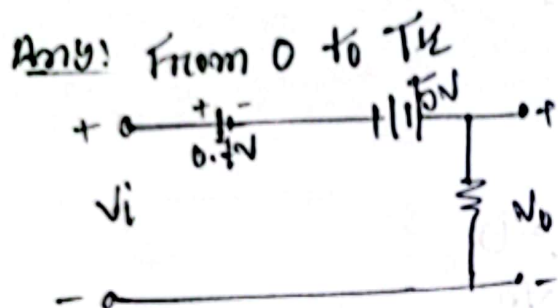
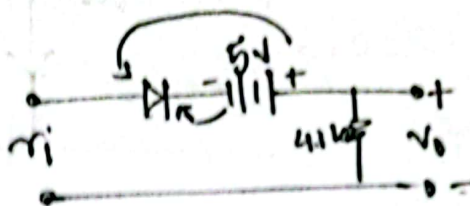
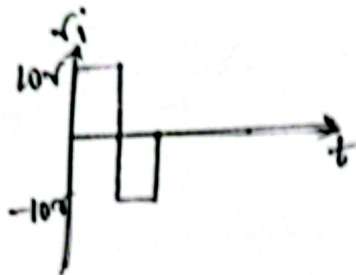
$$v_o = 0V$$

Ans:



Biased Series Clipper:

Determine V_o for the input shown.



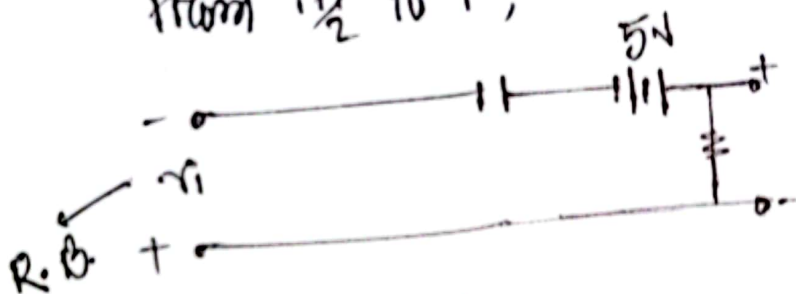
Applying KVL,

$$V_o = V_i - 0.7 + 5V$$

$$= 10 - 0.7 + 5V$$

$$V_o = 14.3V$$

From $T/2$ to T ,

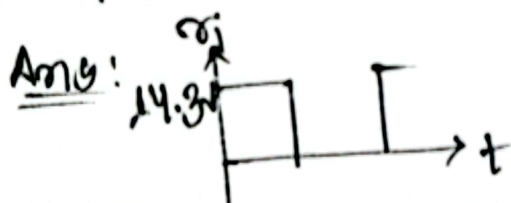


$$V_o = 0V$$

Here, $V_i = 10V$.

Therefore, $V_i > 5V$.

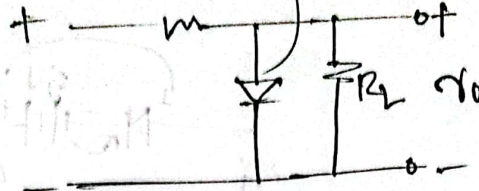
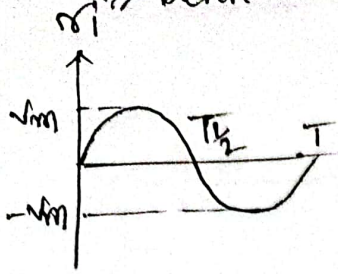
It is not sufficient enough to forward bias.



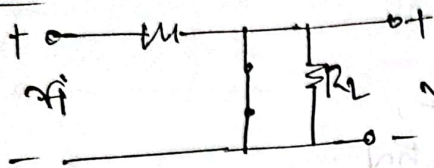
Unbiased Parallel Clipper:

Determine $v_o = ?$

Ideal.

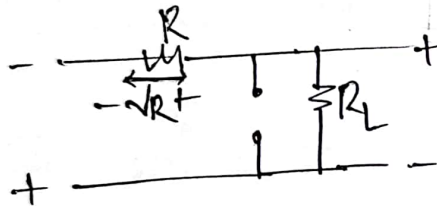


Ans! For 0 to $T/2$,



$$v_o = 0V$$

For $T/2$ to T ,



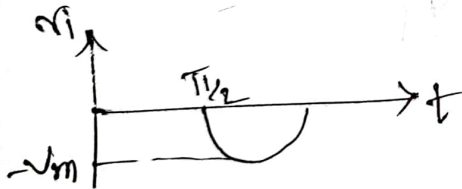
$$v_i + v_o - v_R = 0$$

$$v_o = -v_i + v_R$$

$$\therefore v_o = -v_i$$

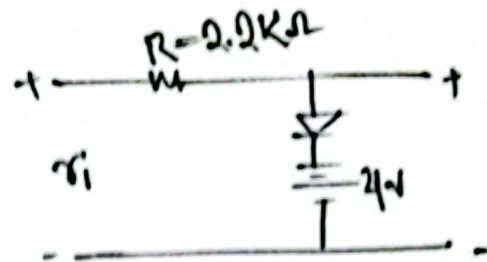
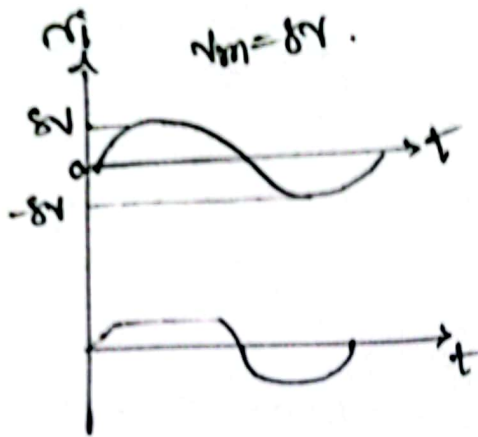
$$v_R \ll$$

Ans!

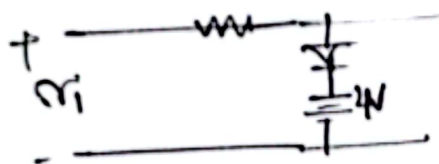


Biased Clipper (Parallel) :

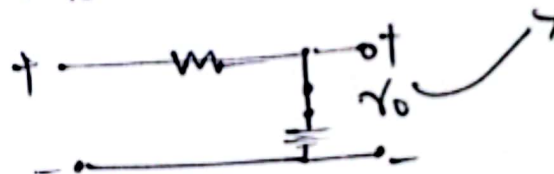
$N_0 = ?$



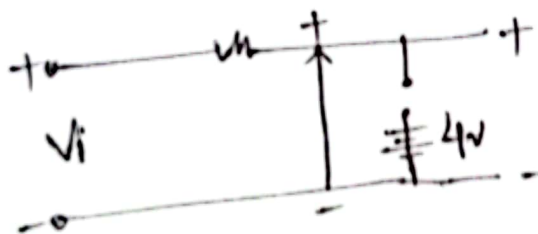
For $0 \text{ to } T_{1/2}$,



F.B $\rightarrow v_i > 4V \therefore \boxed{V_o = 4V}$



For $T_{1/2} \text{ to } T$,



R.B $\rightarrow v_i < 4V$

$V_o = v_i$